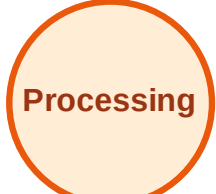


BFS Traversal

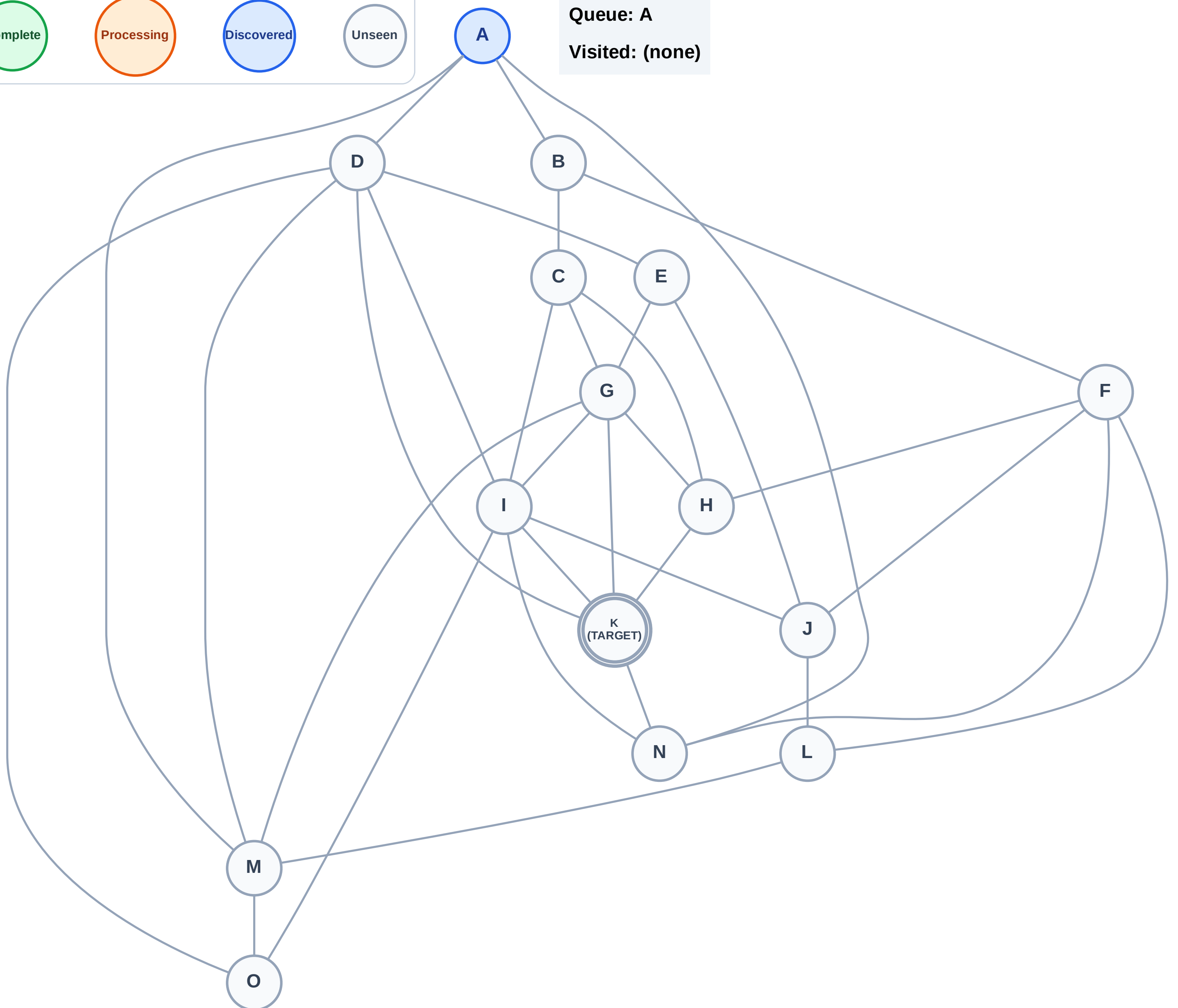
Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend



Queue: A

Visited: (none)



BFS Traversal

Step 2: Process node A

Legend

Complete

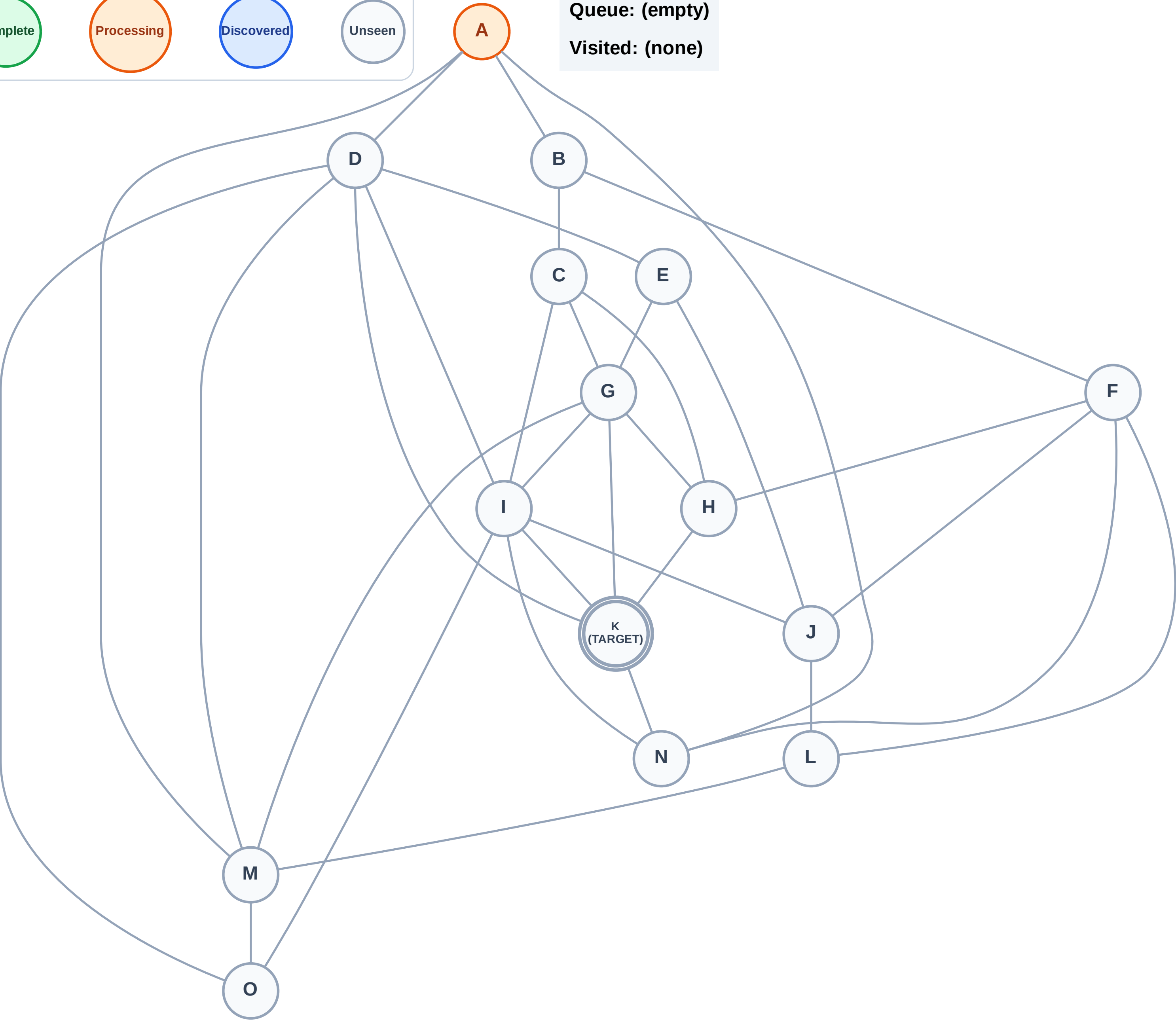
Processing

Discovered

Unseen

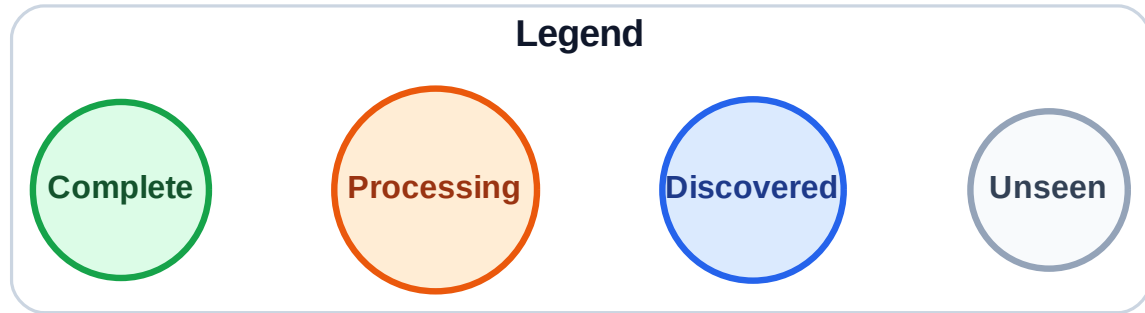
Queue: (empty)

Visited: (none)

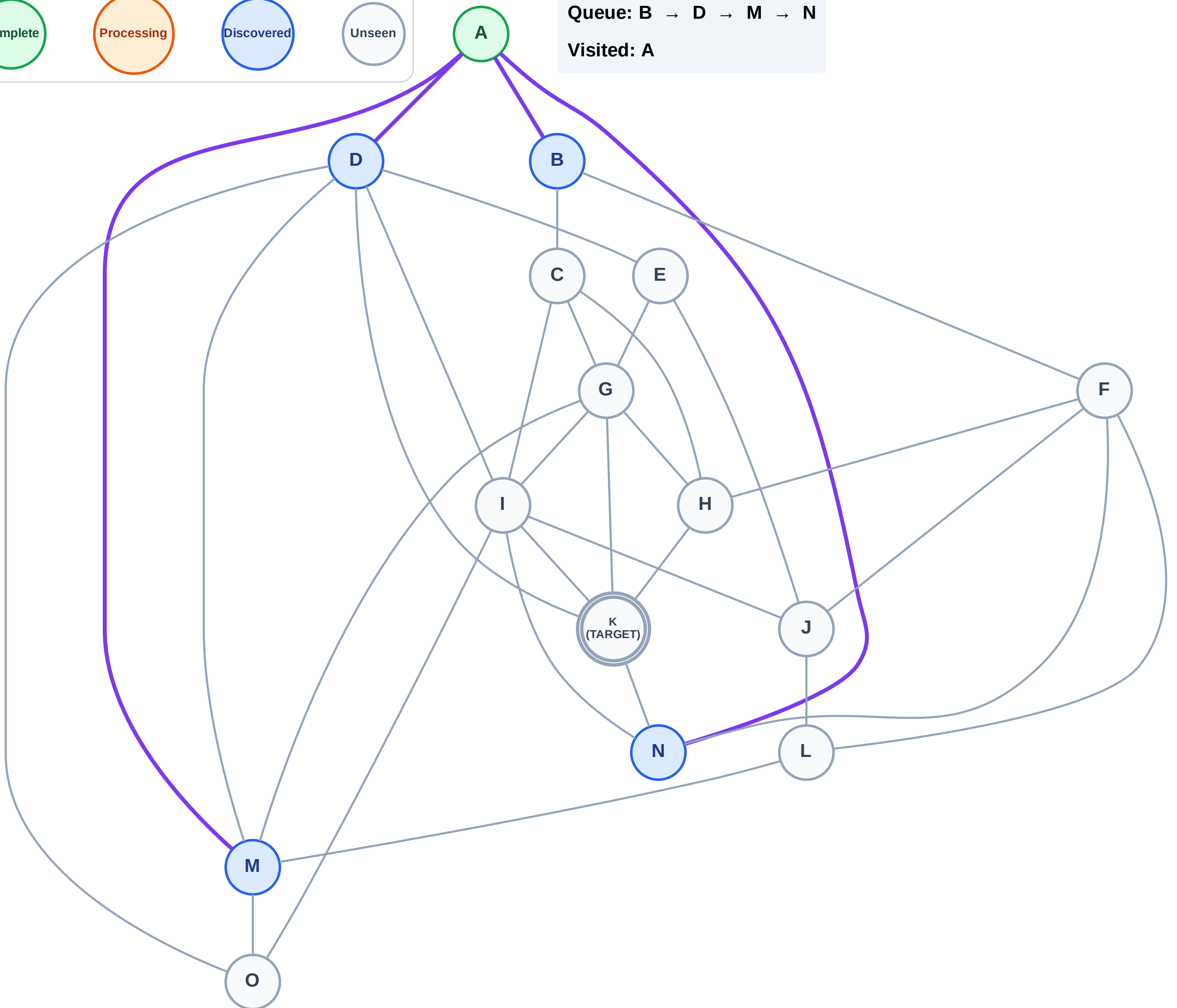


BFS Traversal

Step 3: Discover B, D, M, N from A



Queue: B → D → M → N
Visited: A



BFS Traversal

Step 4: Process node B

Legend

Complete

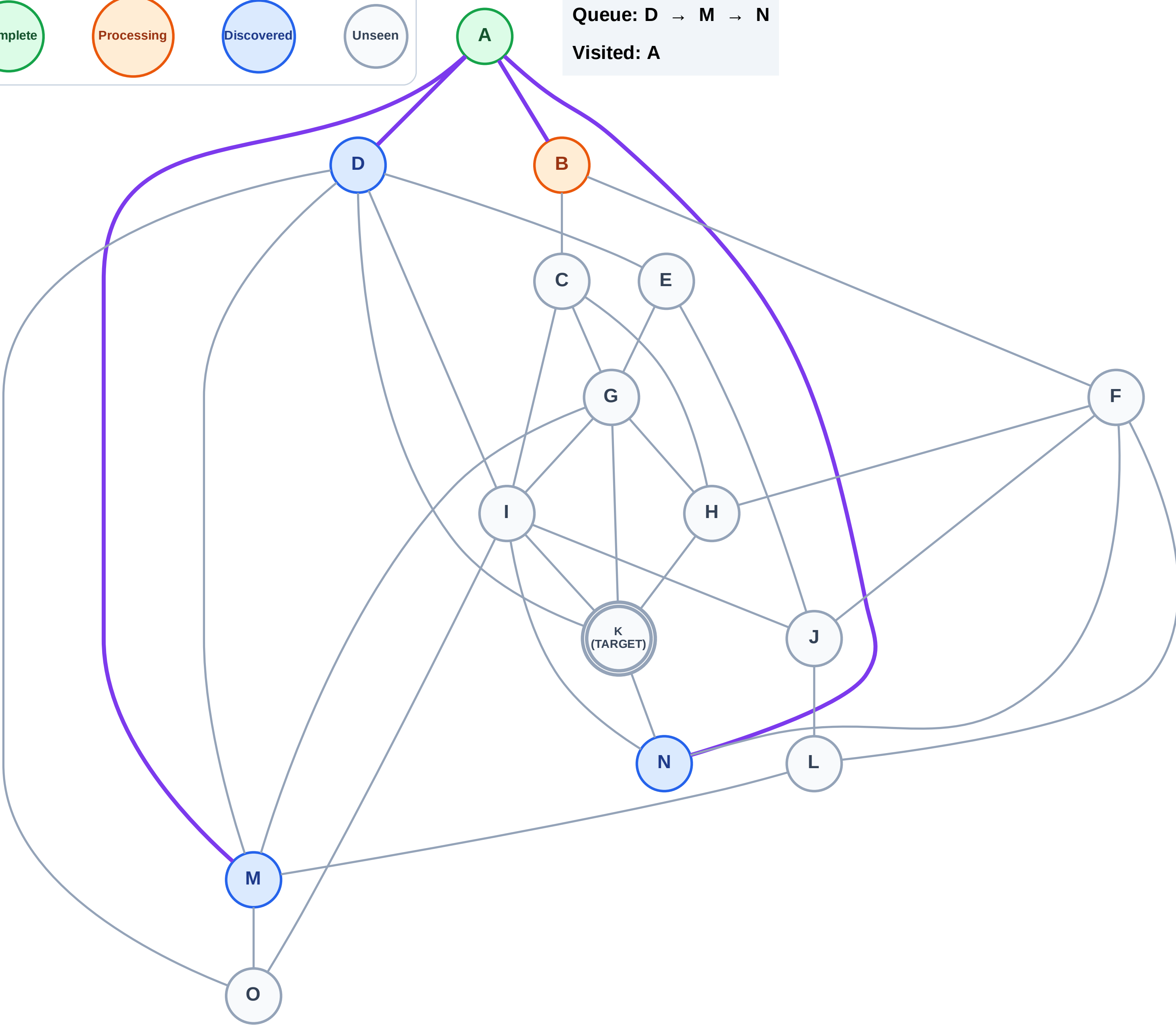
Processing

Discovered

Unseen

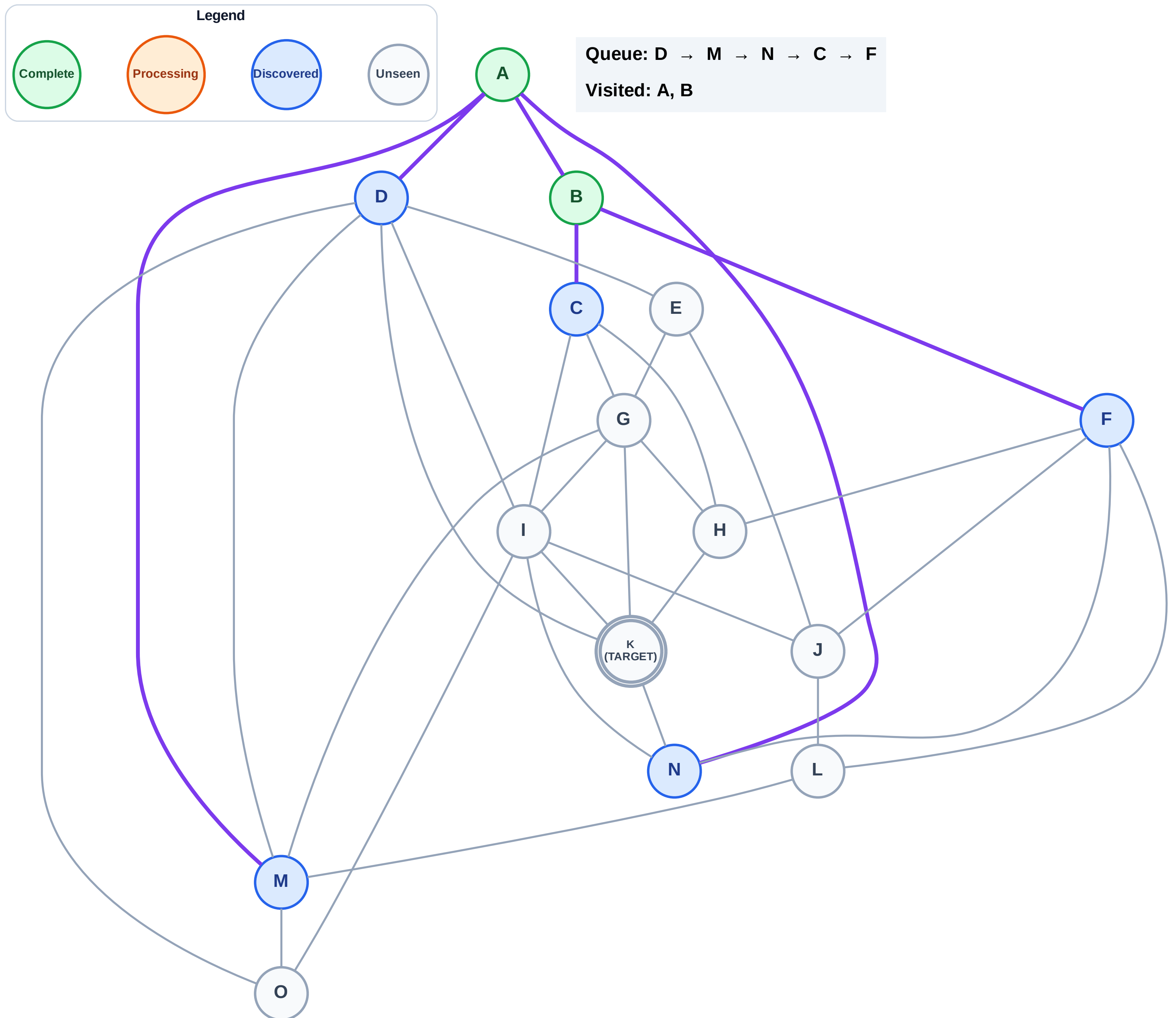
Queue: D → M → N

Visited: A



Step 5: Discover C, F from B

Queue: D → M → N → C → F
Visited: A, B



BFS Traversal

Step 6: Process node D

Legend

Complete

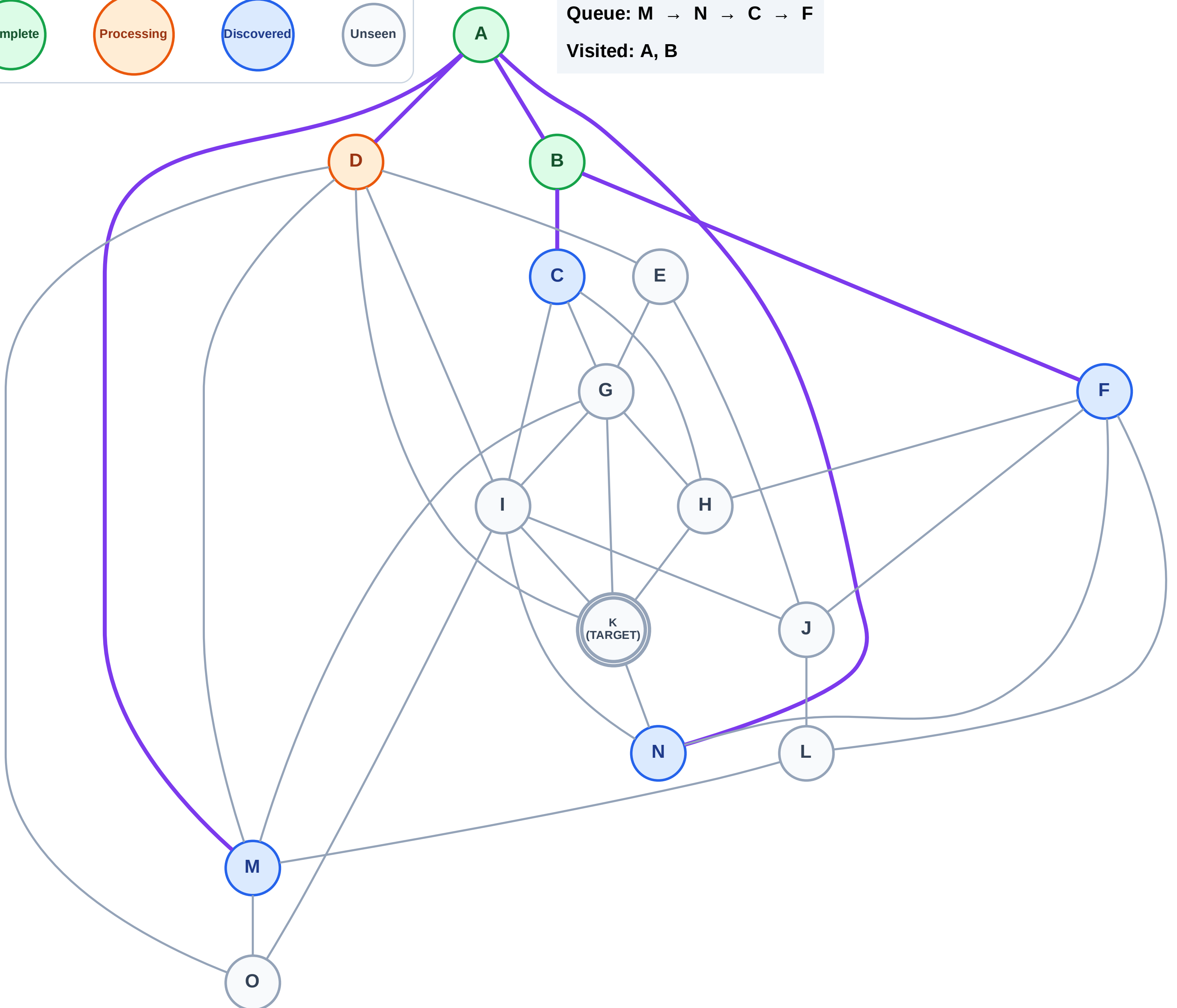
Processing

Discovered

Unseen

Queue: M → N → C → F

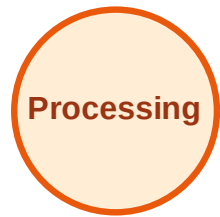
Visited: A, B



BFS Traversal

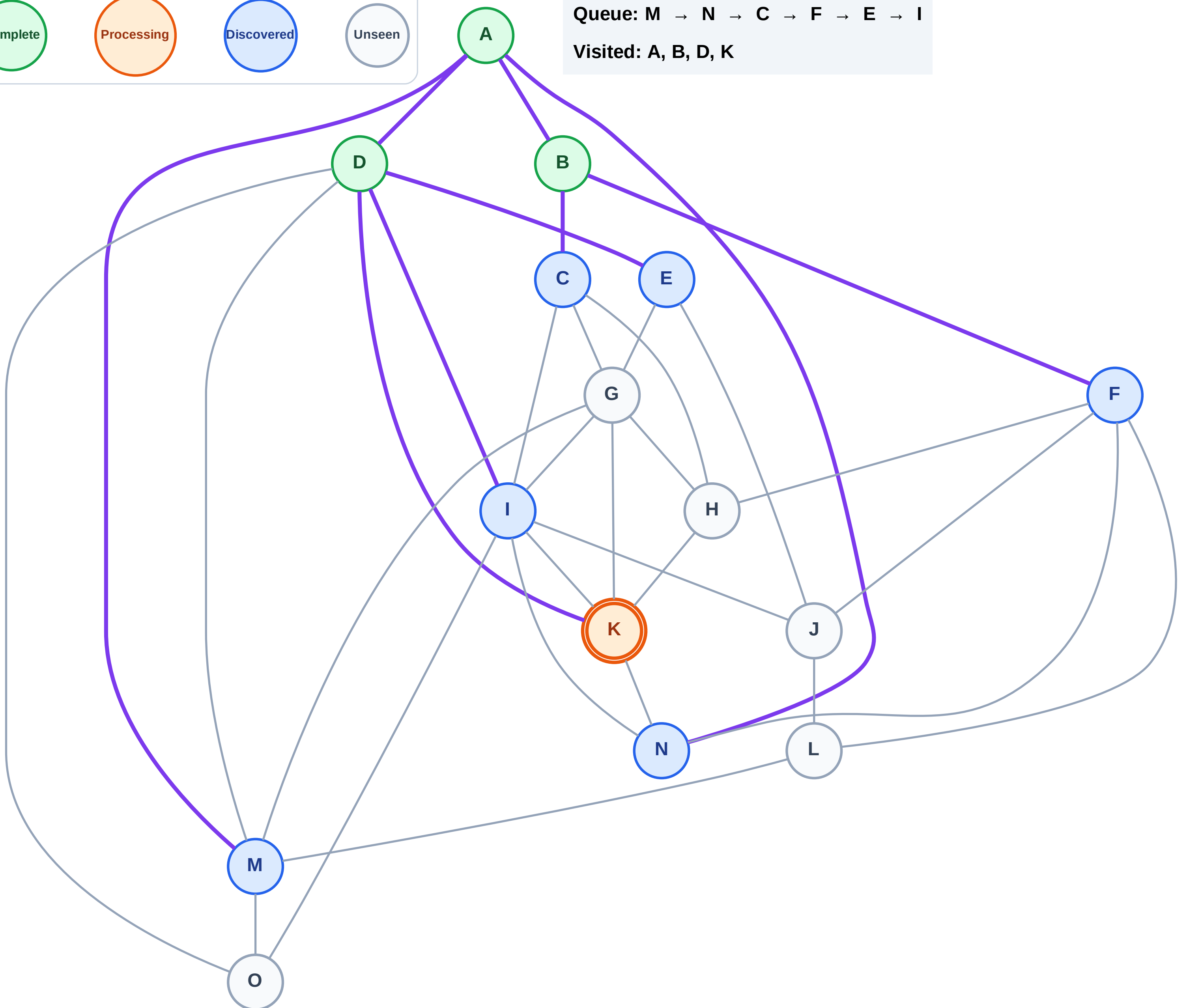
Step 7: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (2 hops): A → D → K

Legend



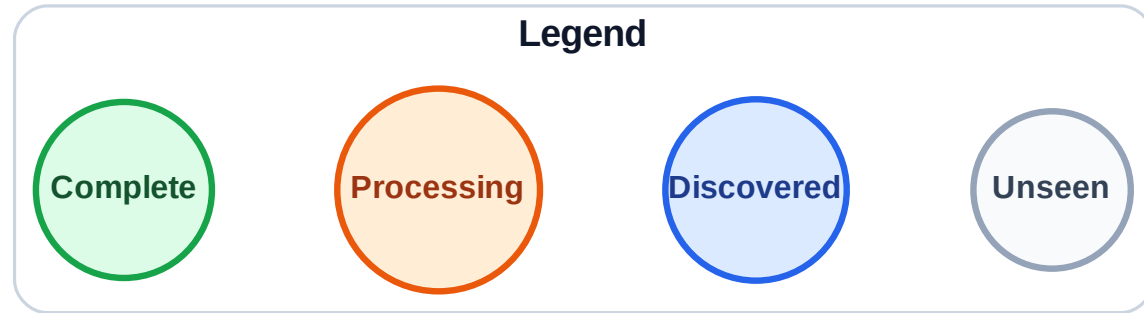
Queue: M → N → C → F → E → I

Visited: A, B, D, K

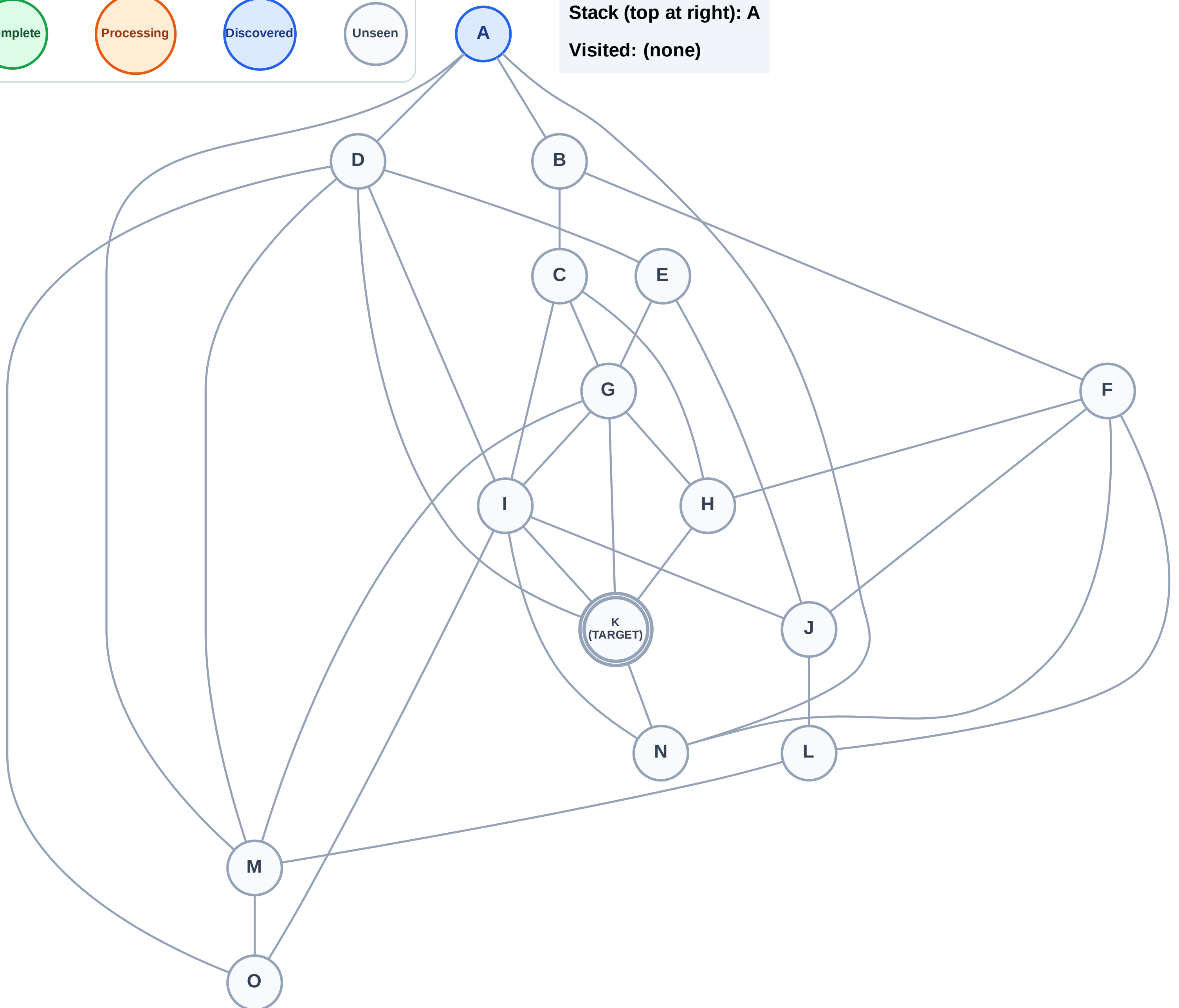


DFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))



Stack (top at right): A
Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

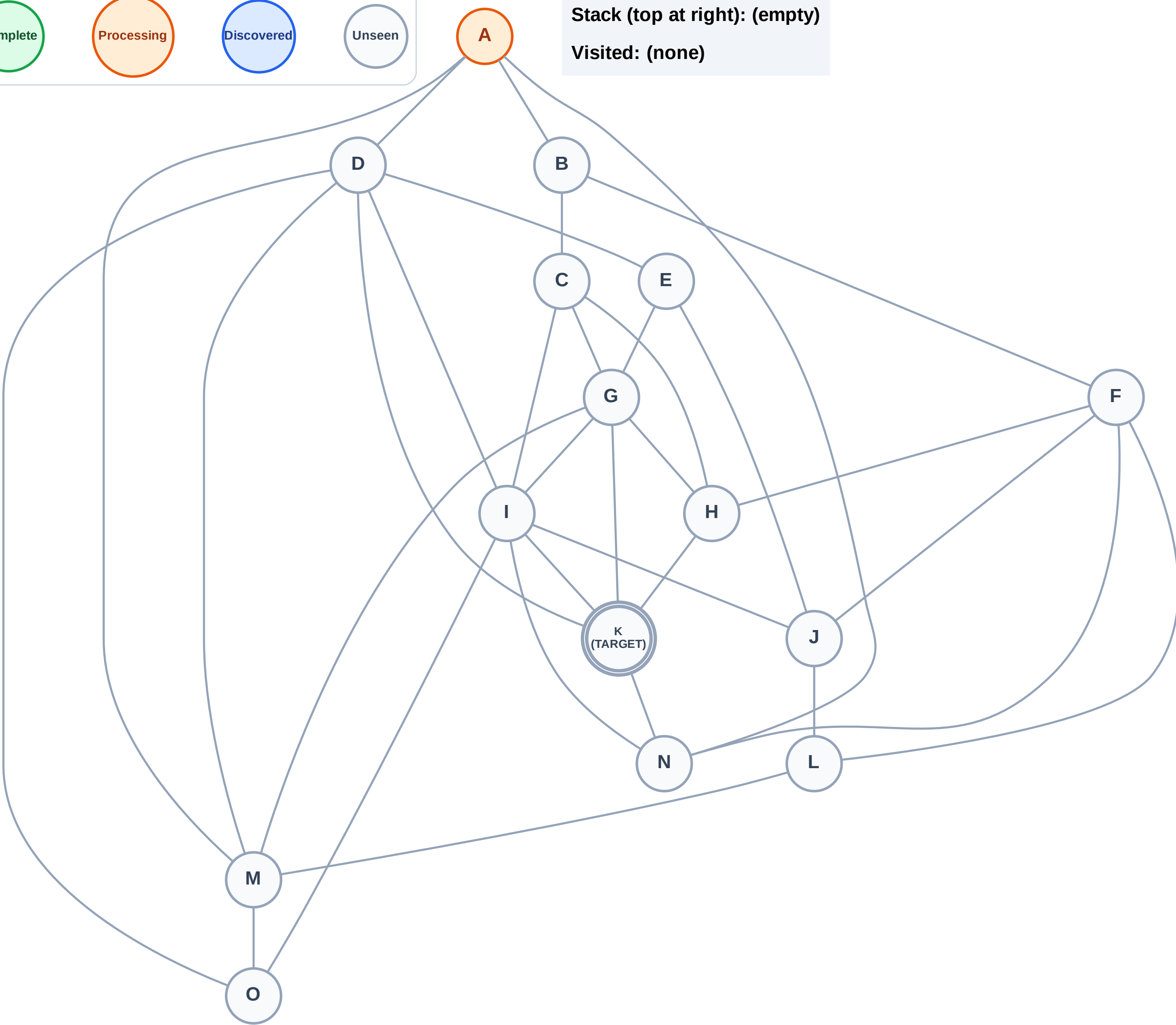
Complete

Processing

Discovered

Unseen

Stack (top at right): (empty)
Visited: (none)



DFS Traversal

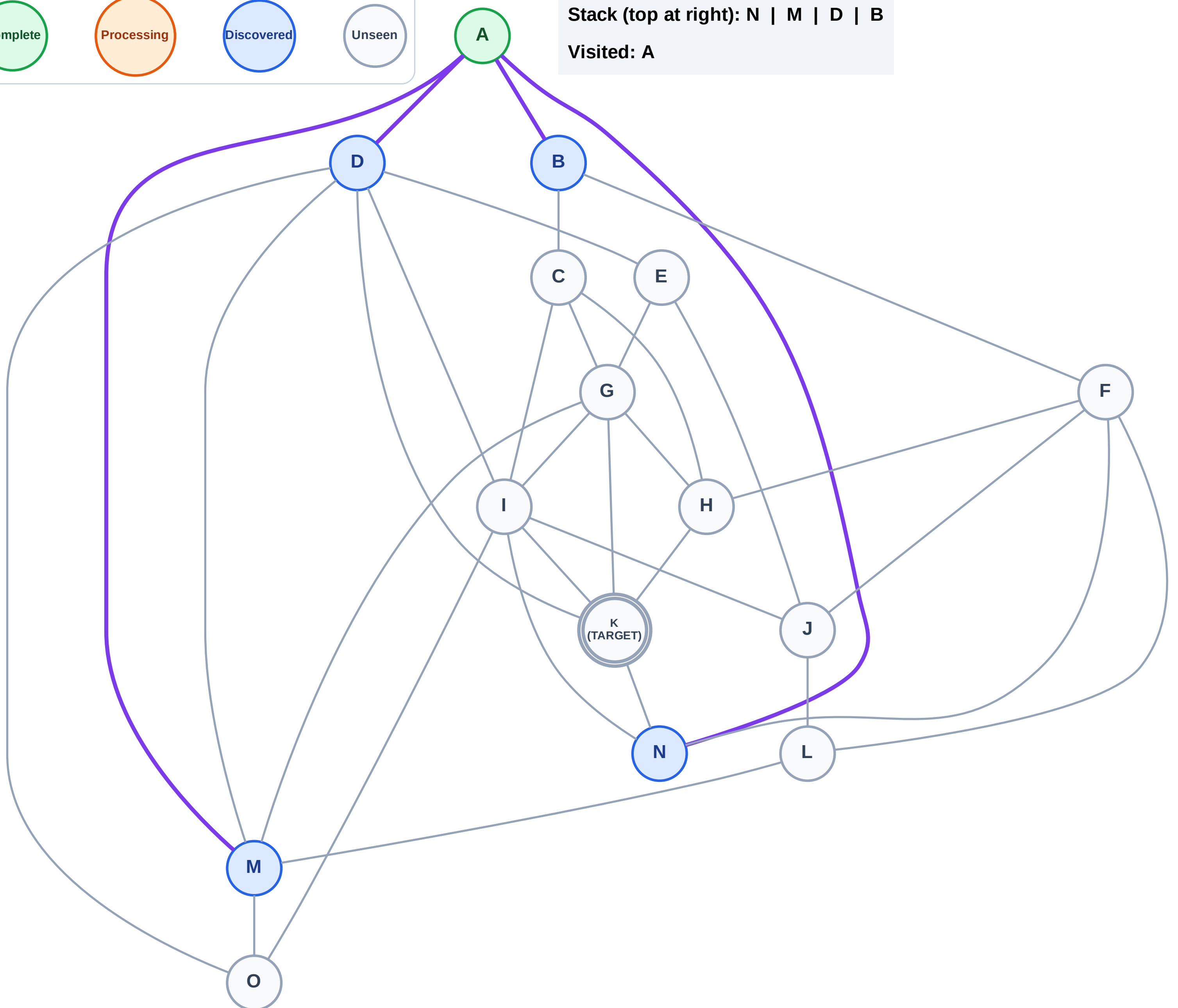
Step 3: Discover N, M, D, B from A

Legend



Stack (top at right): N | M | D | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

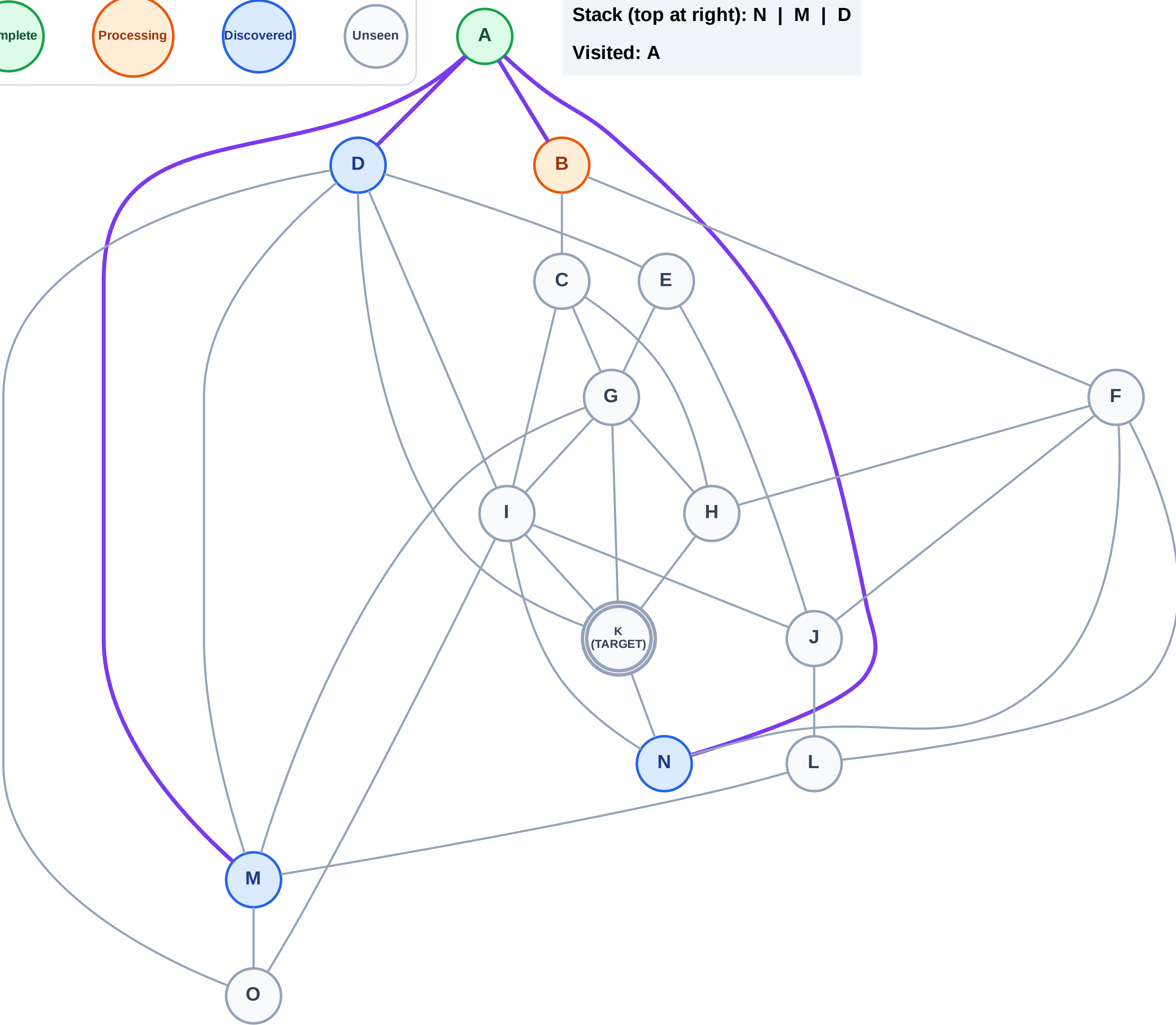
Complete

Processing

Discovered

Unseen

Stack (top at right): N | M | D
Visited: A



DFS Traversal

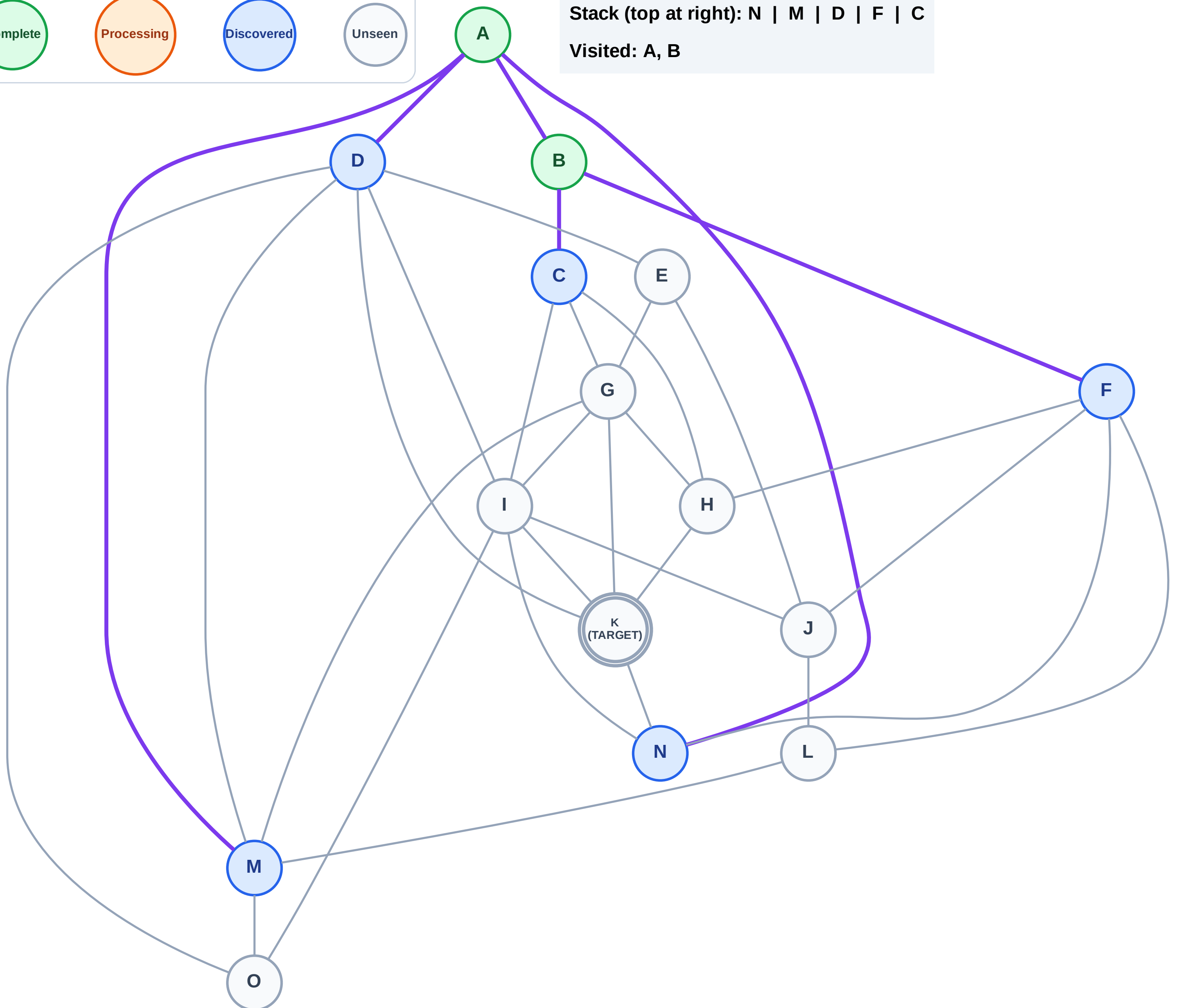
Step 5: Discover F, C from B

Legend



Stack (top at right): N | M | D | F | C

Visited: A, B



DFS Traversal

Step 6: Process node C

Legend

Complete

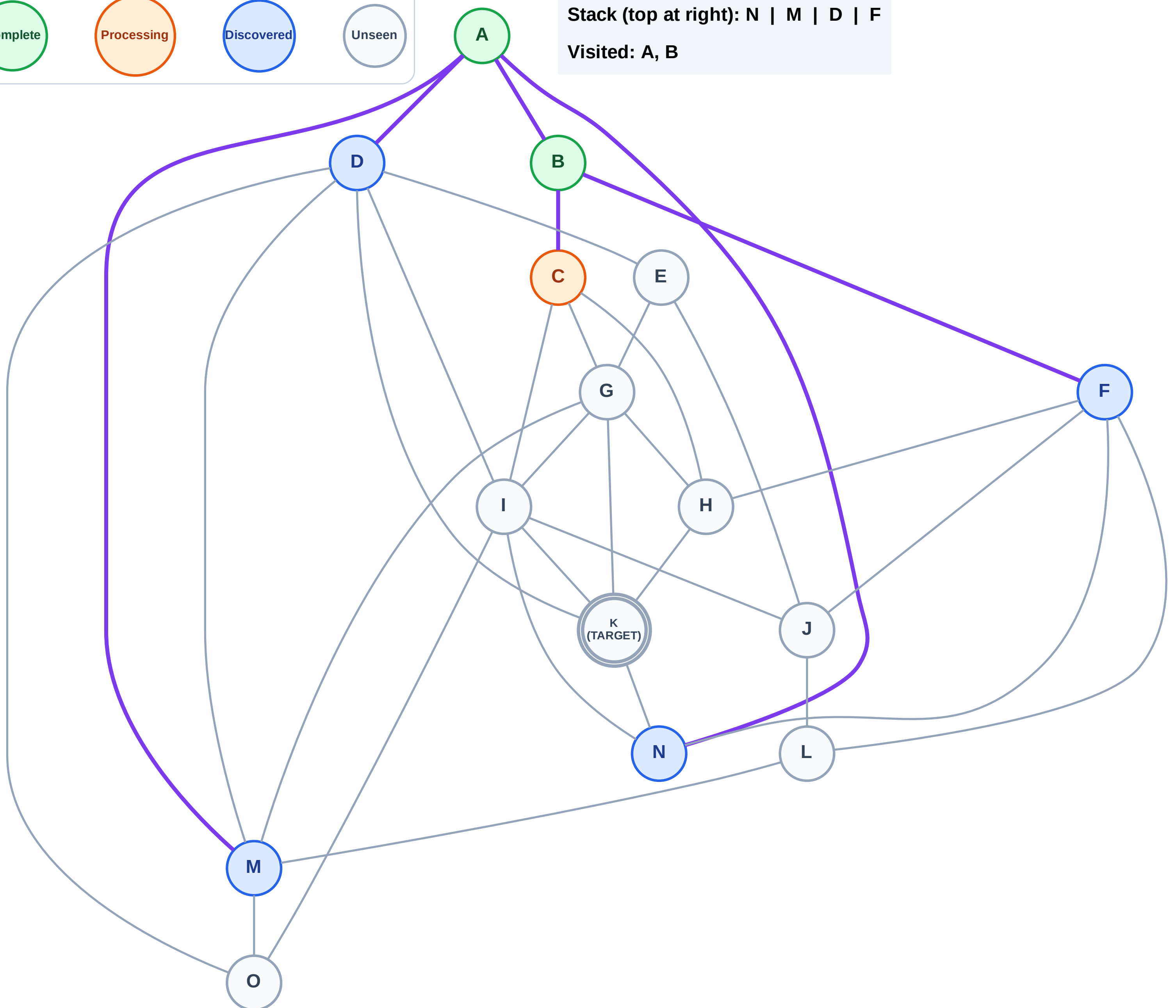
Processing

Discovered

Unseen

Stack (top at right): N | M | D | F

Visited: A, B



DFS Traversal

Step 7: Discover I, H, G from C

Legend

Complete

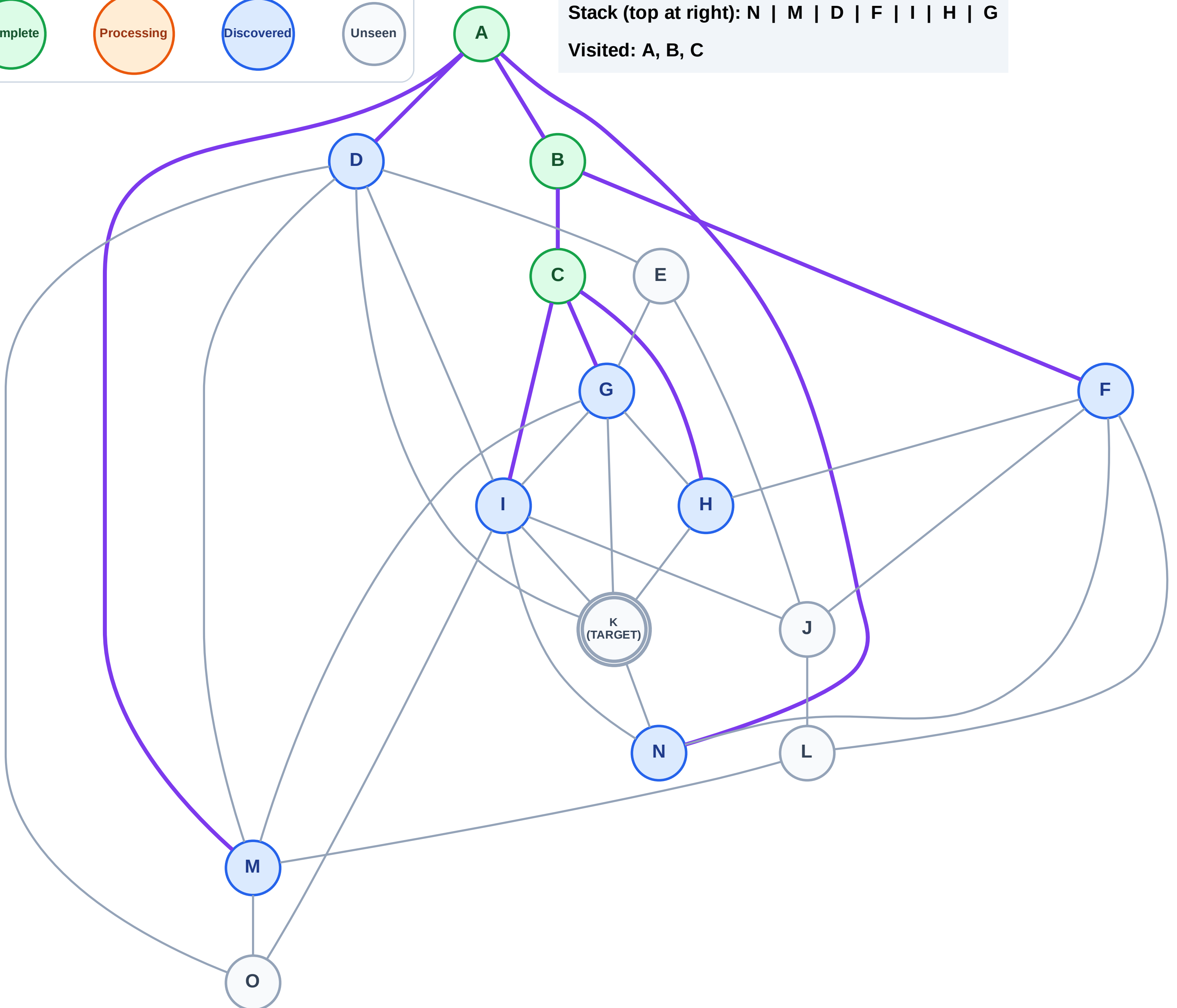
Processing

Discovered

Unseen

Stack (top at right): N | M | D | F | I | H | G

Visited: A, B, C



DFS Traversal

Step 8: Process node G

Legend

Complete

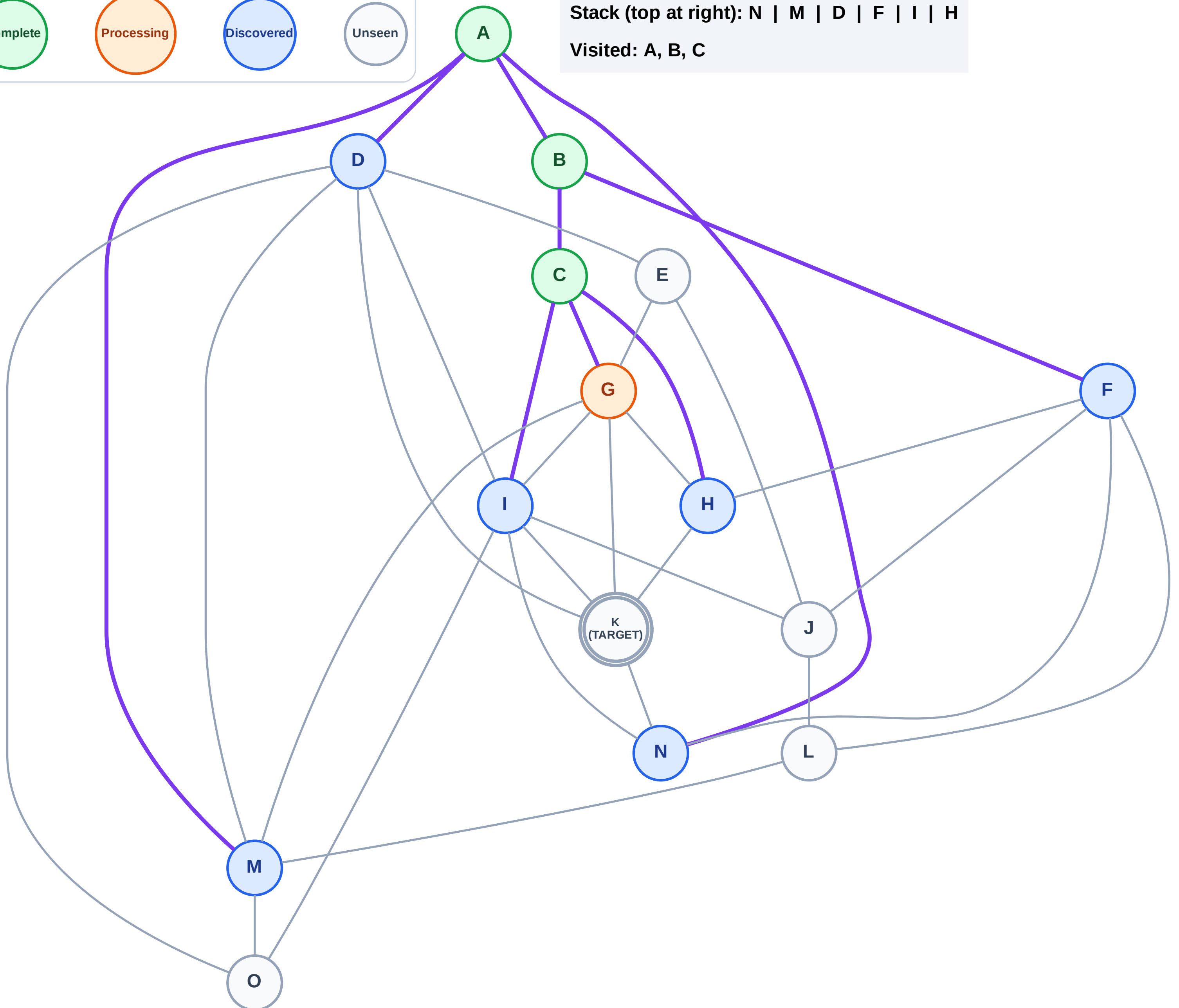
Processing

Discovered

Unseen

Stack (top at right): N | M | D | F | I | H

Visited: A, B, C



DFS Traversal

Step 9: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (4 hops): A → B → C → G → K

Legend

Complete

Processing

Discovered

Unseen

Stack (top at right): N | M | D | F | I | H
Visited: A, B, C, G, K

